Automatic Solving

Oussama Belhout, M1 Informatique

Scheduling Challenges

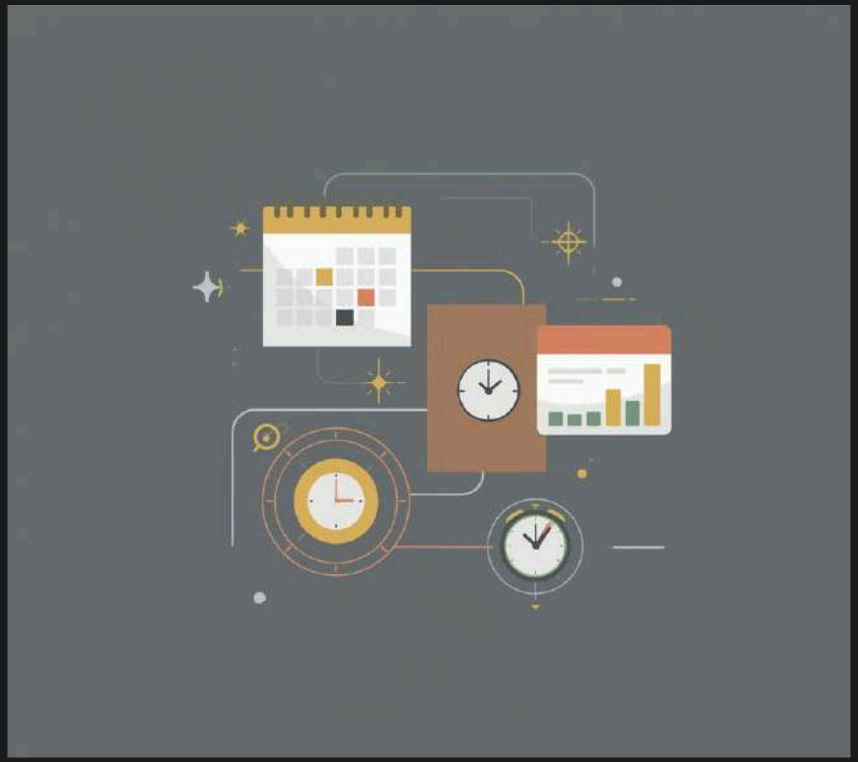
How do we effectively allocate 20 nurses across 7 shifts daily?

Scheduling nurses involves complex decision-making to ensure adequate coverage and **patient care**.

Common Structures

Scheduling

Efficiently organizing tasks and resources is crucial.



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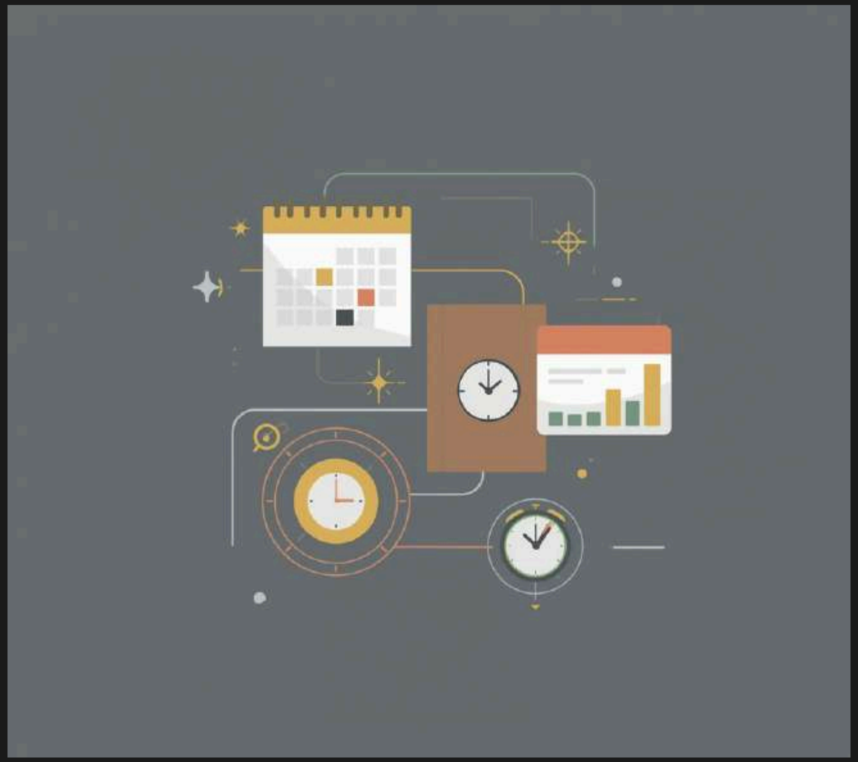
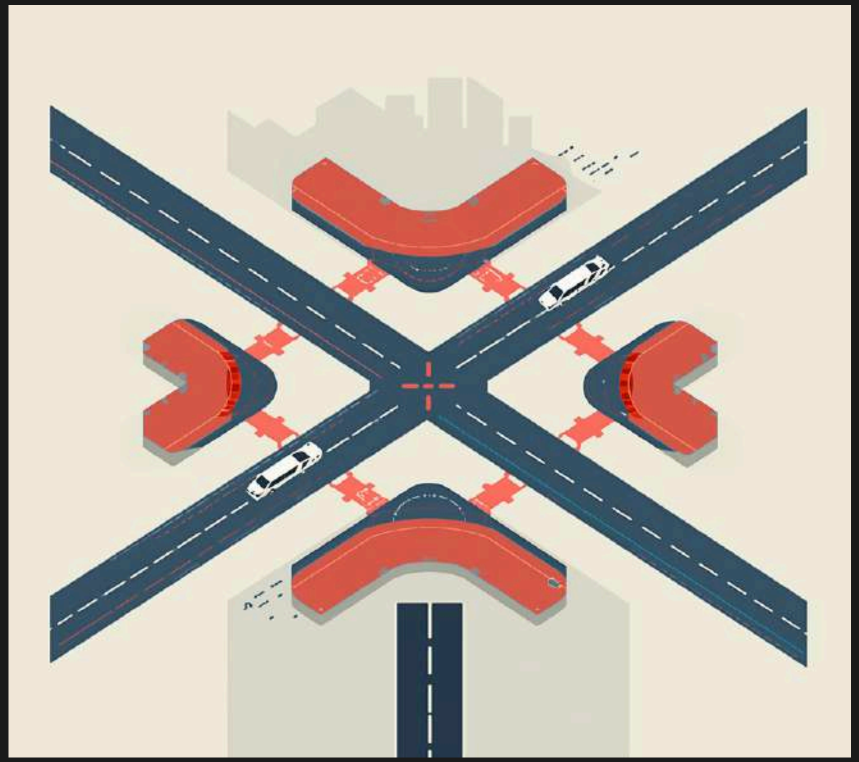


Routing

Navigating pathways optimizes delivery and logistics.



Common Structures

Scheduling

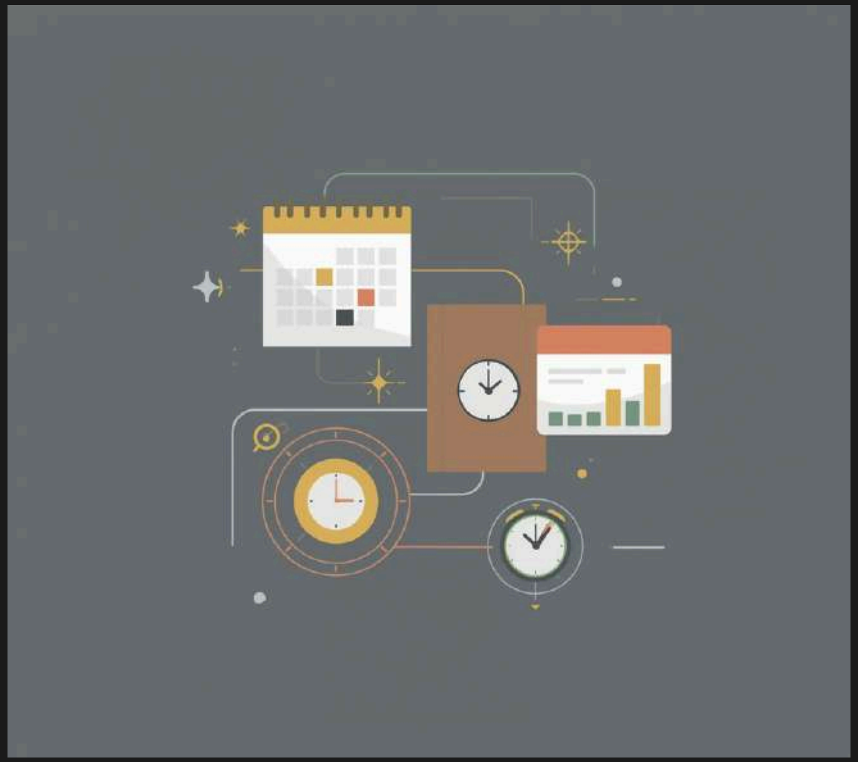
Efficiently organizing tasks and resources is crucial.

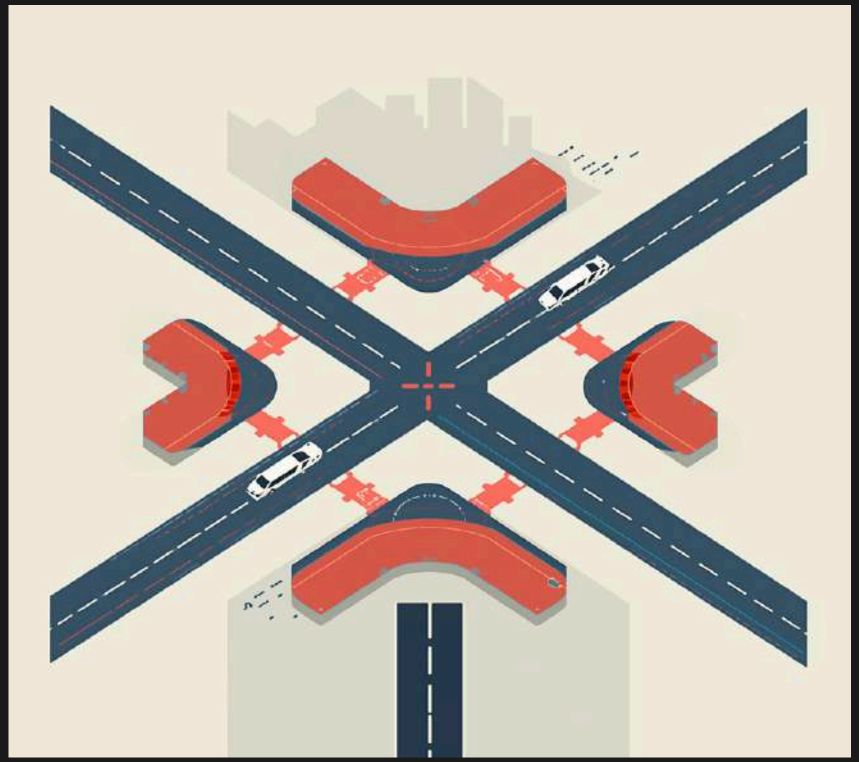
Routing

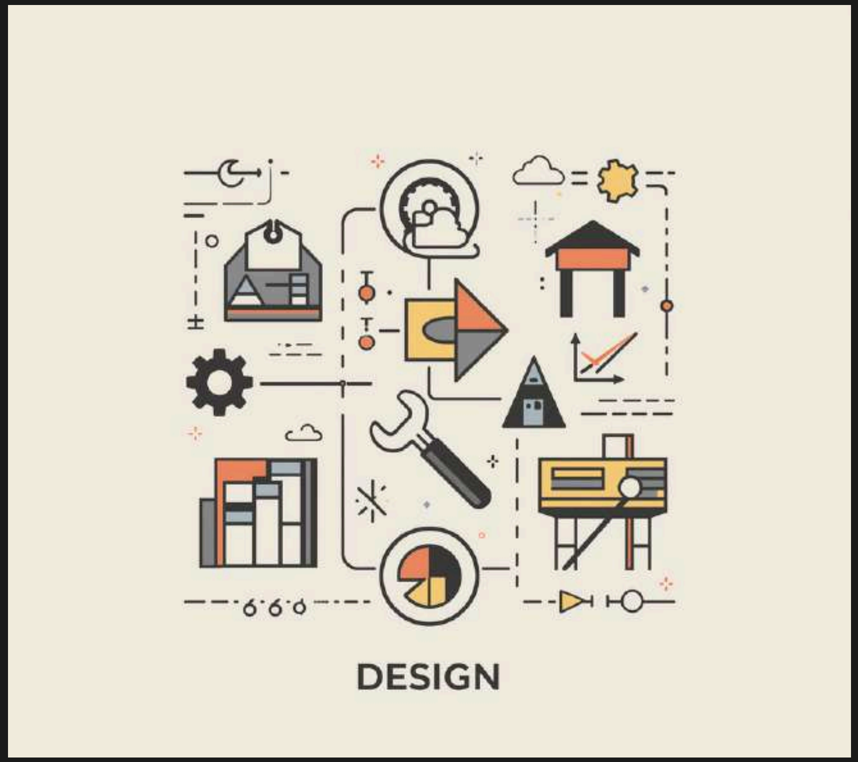
Navigating pathways optimizes delivery and logistics.

Designing

Creating layouts improves functionality and performance.



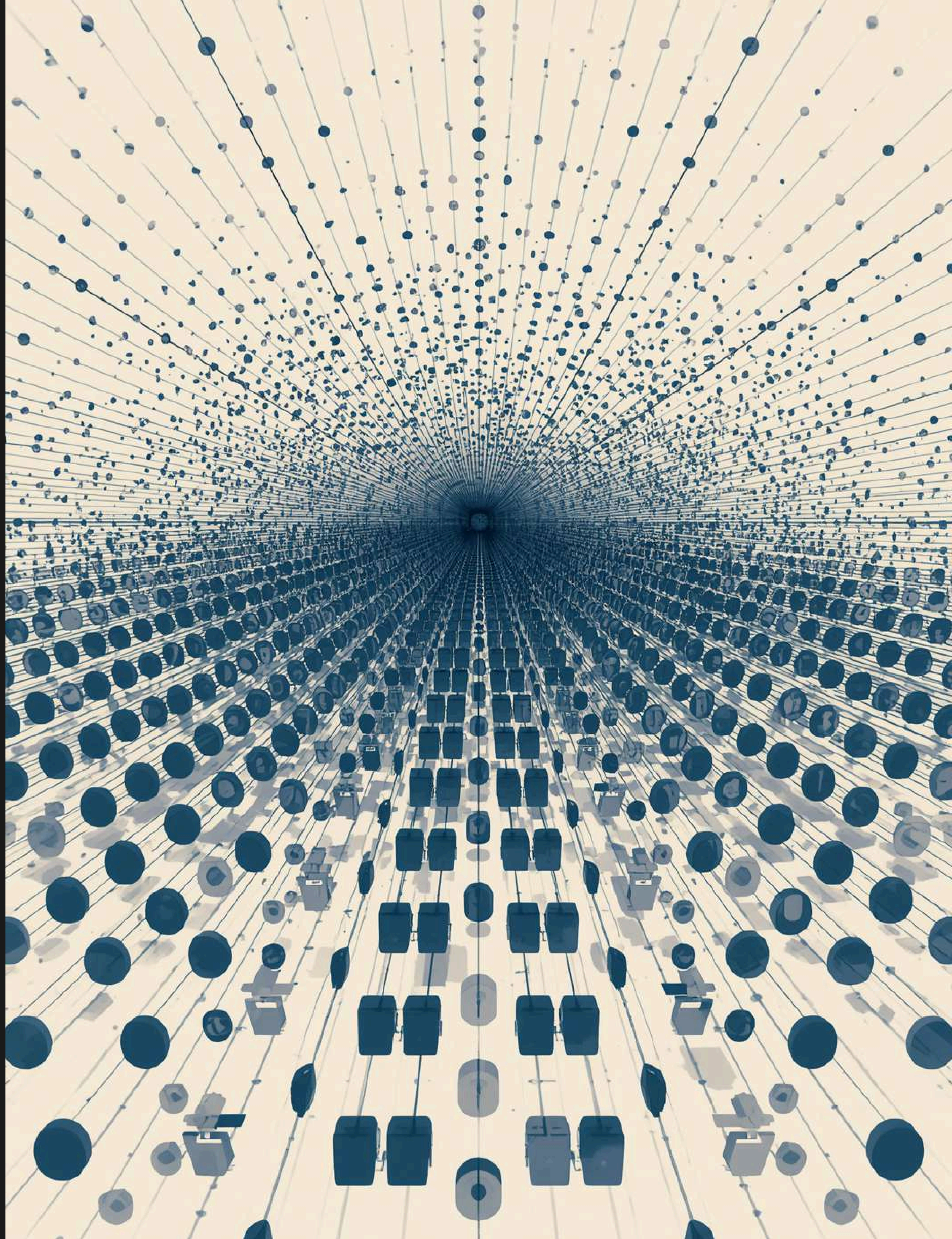




The Challenge of Scale

Understanding the vastness of possible scheduling assignments for employees

Scheduling **20 employees** for 7 shifts presents an overwhelming challenge, resulting in more than **10^20 possible assignments**.



Meet the Solver: The Search for Solutions

The Search Process

This solver efficiently navigates vast possibilities to find solutions.

Solvers' Language

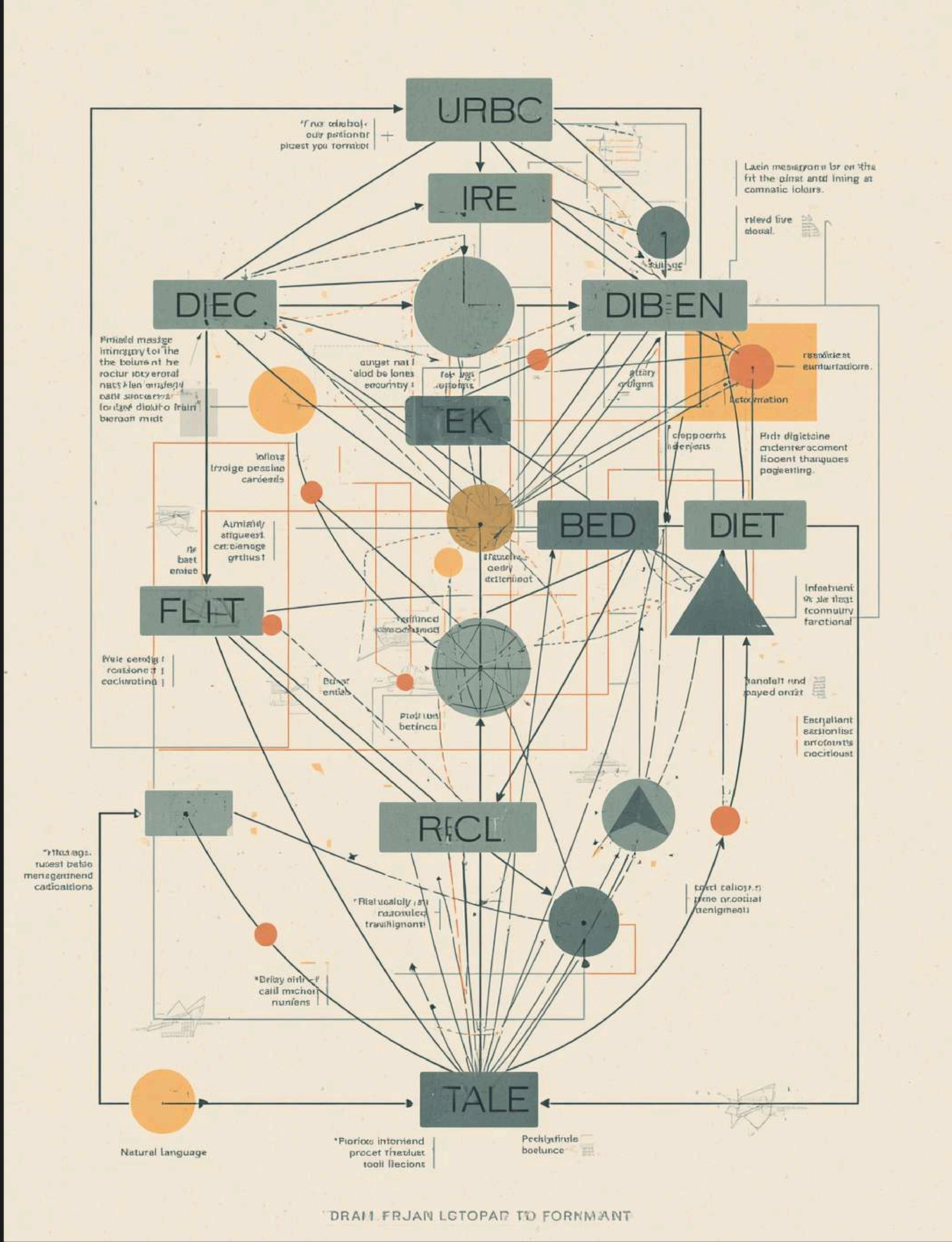
Bridging the Gap Between Human Description and Formal Models

Solvers require **specific formal models** to operate effectively, which poses a significant barrier for non-specialists.

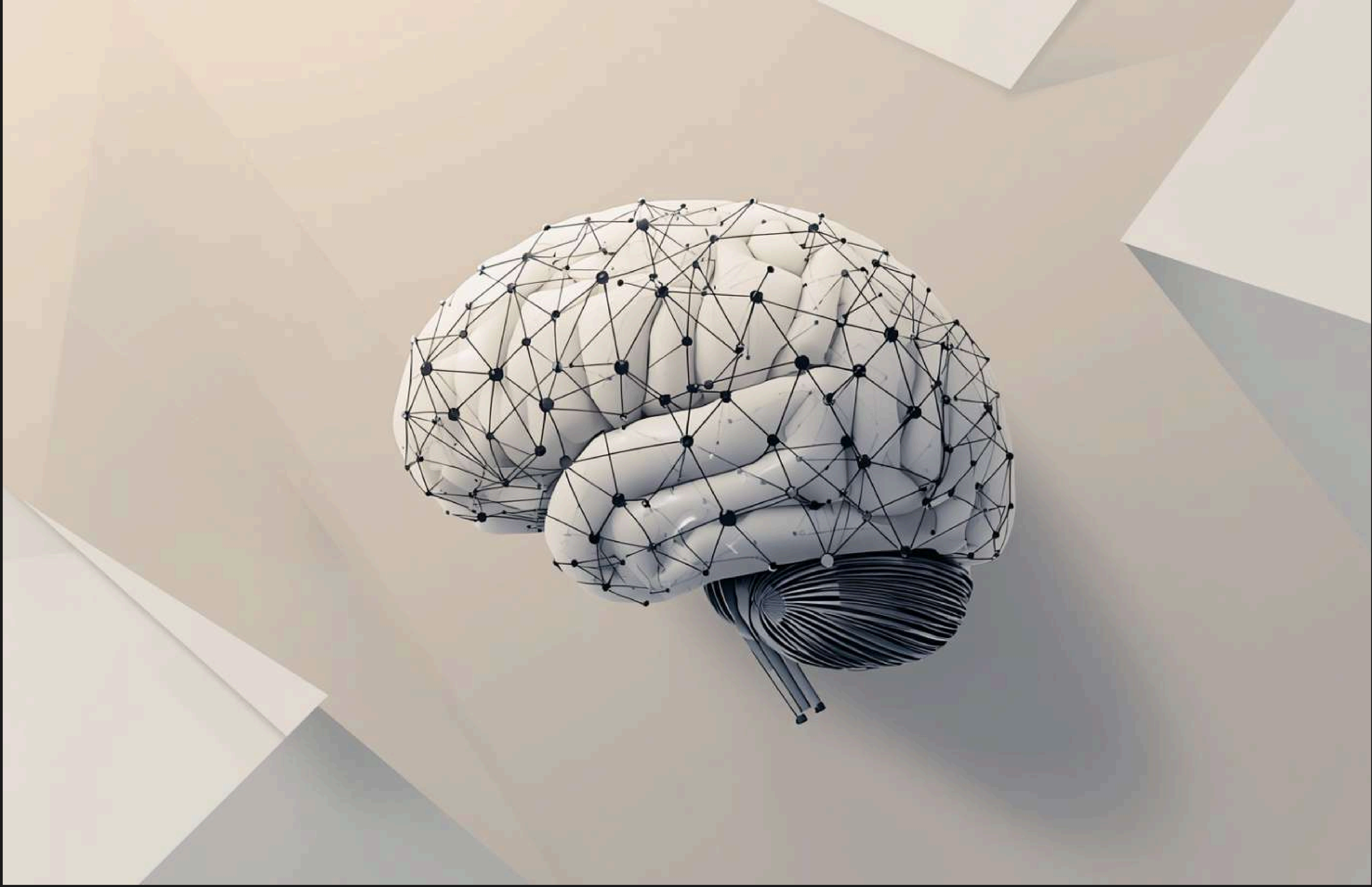
The Translation Challenge

Automating formal model creation is crucial for leveraging LLMs effectively

Effective automation requires **expertise in formal modeling**, which is often a barrier for many practitioners.



Enter the Large Language Model



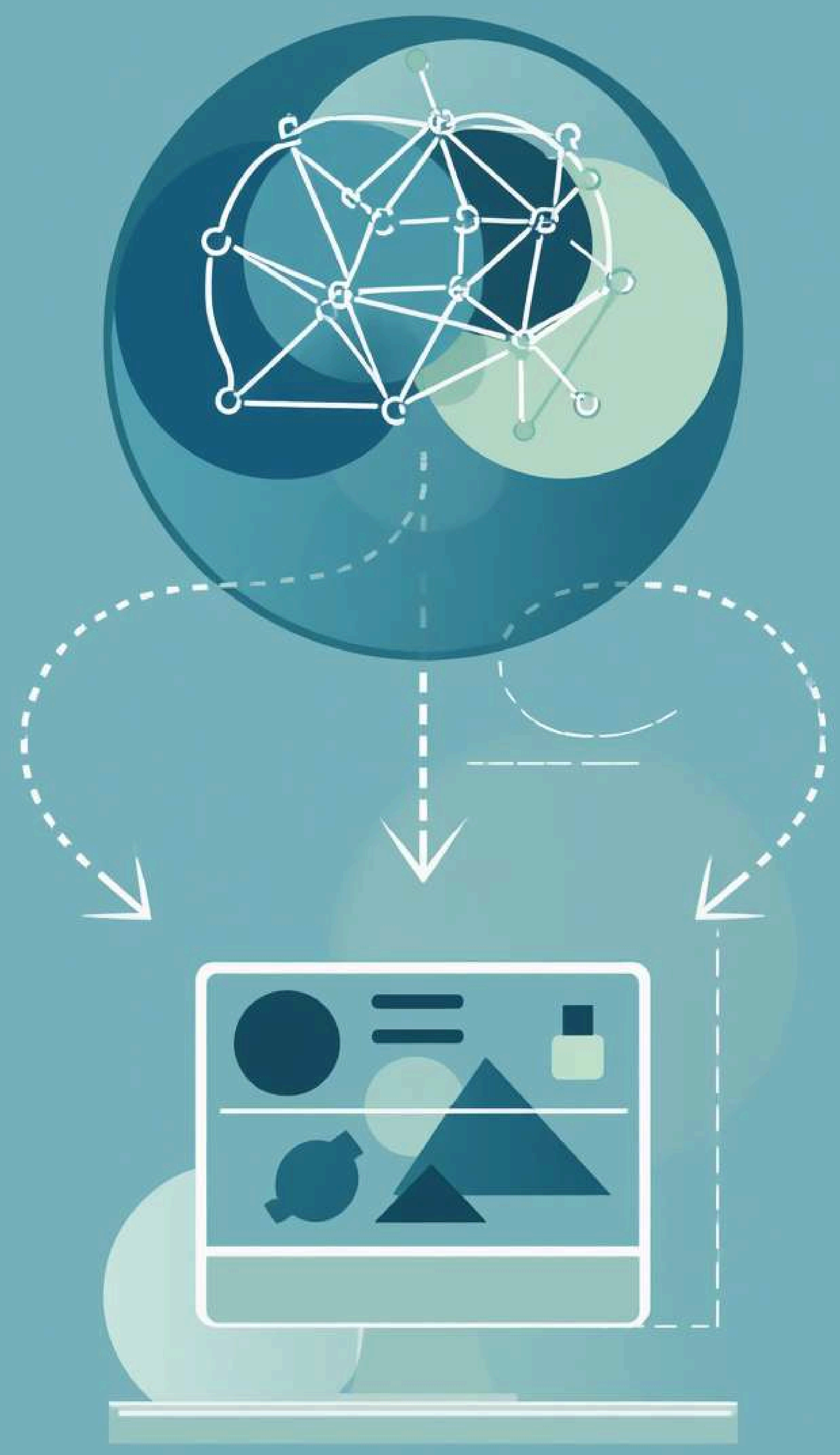
LLM Capabilities

An LLM understands language and translates it into code.

Core Idea

The LLM translates while the solver computes the solution efficiently

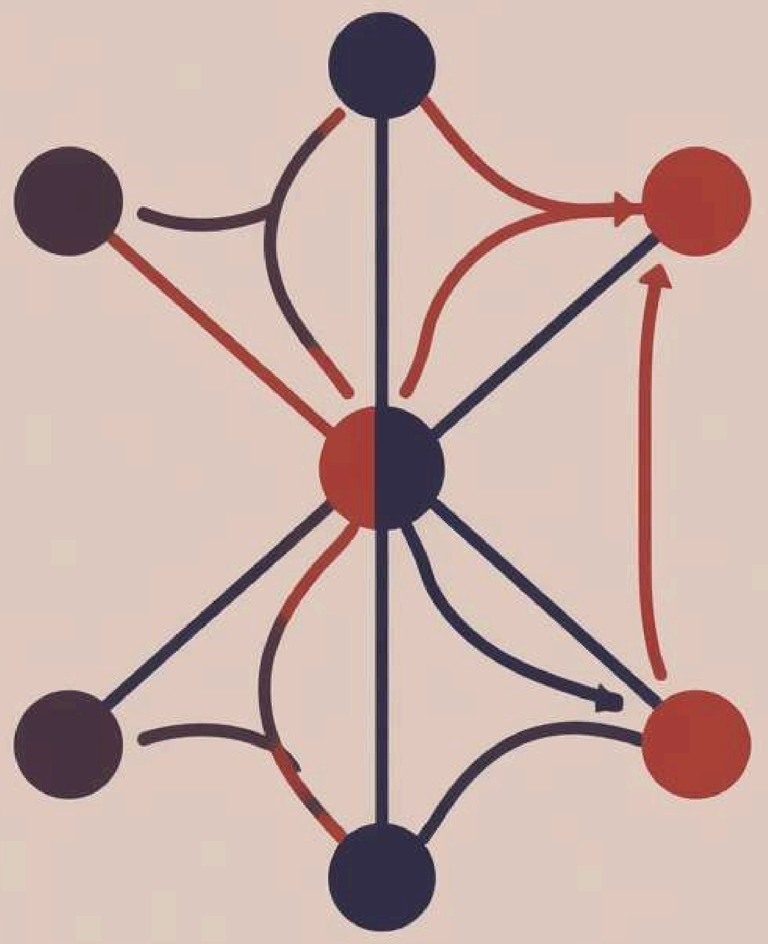
This work explores a **collaborative framework** where a Large Language Model interprets natural language inputs and formulates them into formal models.



The Perfect Match

Combining creativity and precision in problem-solving for combinatorial challenges

The integration of an LLM and a solver creates a **dynamic partnership**. The LLM offers flexibility and creativity in interpreting problems, while the solver ensures accuracy and reliability in execution.



A Concrete Example: The 4 Queens Problem

Solution

The queens are positioned to **avoid each other’s attacks**.

Understanding CSP Elements

Explore the core components of CSPs

Core Components

- Variables
- Domains
- Constraints
- Assignments

Key Concepts

- Value Selection
- Rule Adherence
- Solution Space
- Problem Solving

Design Choices

Prompting Technique

Effective phrasing impacts LLM's performance.

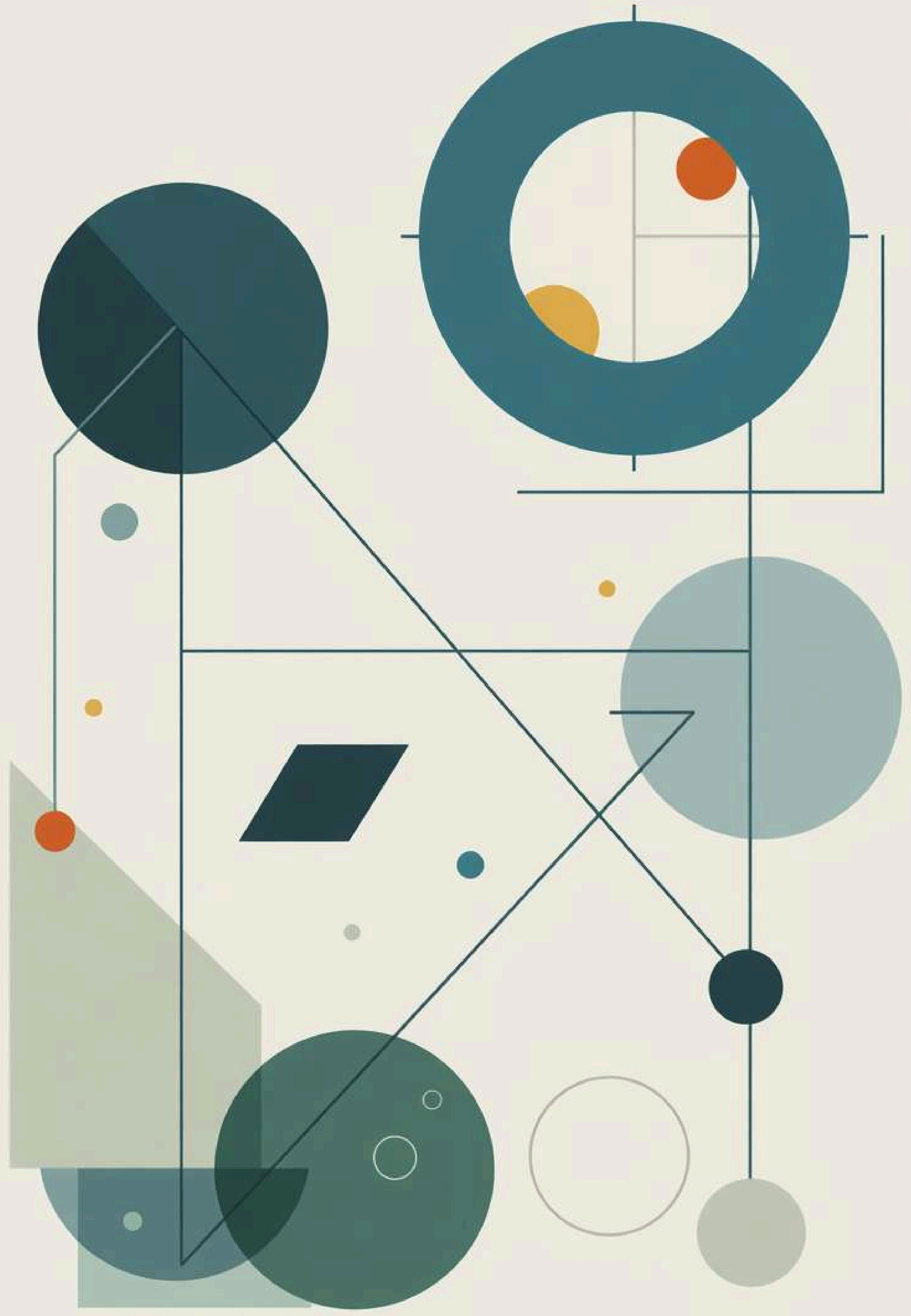
LLM Model

Different models offer unique strengths for tasks.

Prompting Technique

The art of crafting effective requests for LLMs

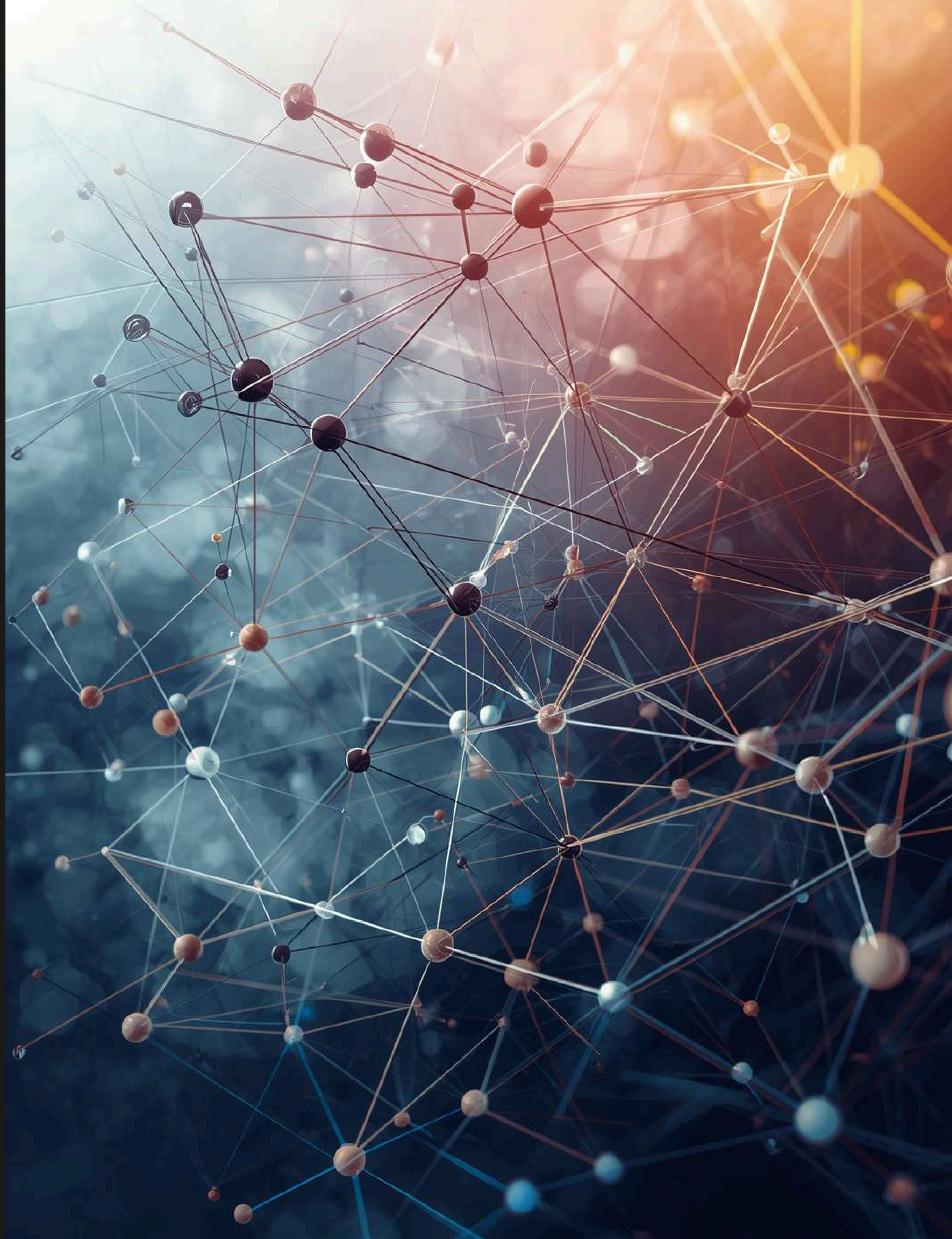
The way we phrase our requests significantly influences the responses from LLMs. **Effective prompting** is essential for obtaining accurate results. By utilizing various prompting techniques, such as few-shot or chain-of-thought approaches, we can guide the LLM to better understand the task and improve overall performance.



The Model

Exploring the diverse capabilities of various LLMs for combinatorial problem-solving

Different LLMs exhibit unique strengths that affect their performance in solving combinatorial problems. By testing models like Qwen, Claude, and Haiku, we can determine which excels in specific tasks and how their architectures influence outcomes. This understanding is crucial for optimizing our approach to complex problem-solving.



Agentic Design Choices

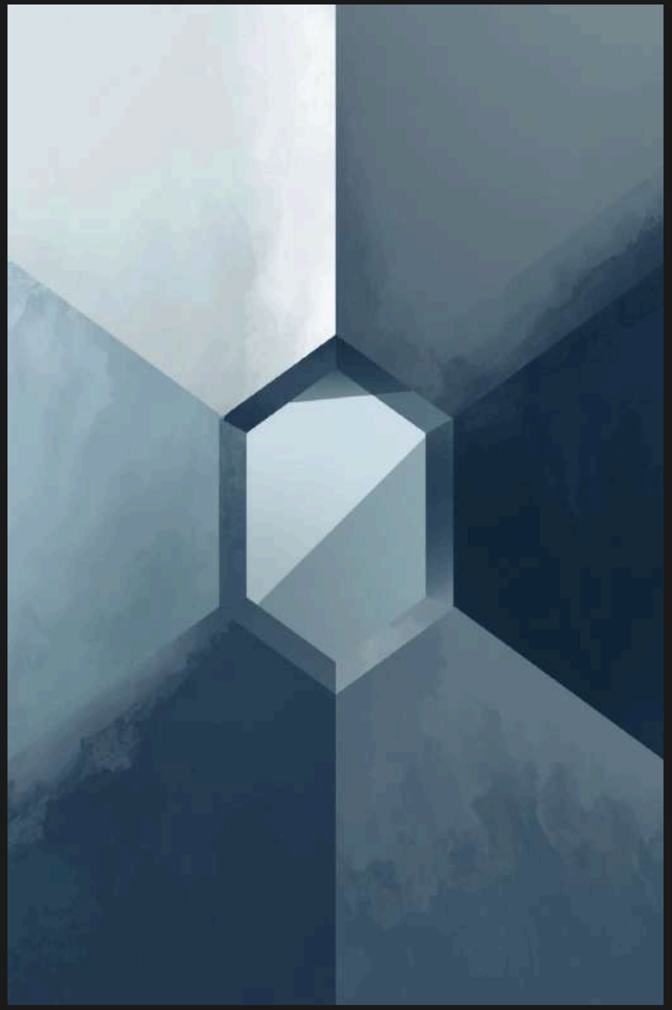
Evaluating the optimal number of specialists for effective problem-solving

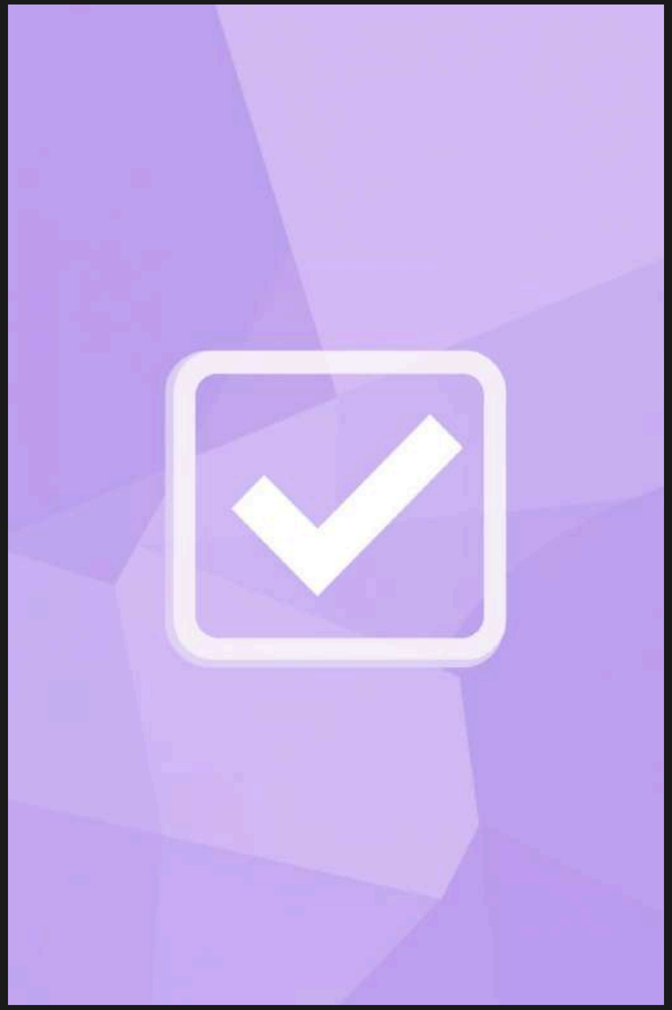
The **design of our system** hinges on the number of specialists involved. Hiring too few may limit exploration, while too many can introduce **communication overhead**. Striking the right balance is crucial for achieving efficiency and maximizing the solver's capability to tackle complex combinatorial problems.

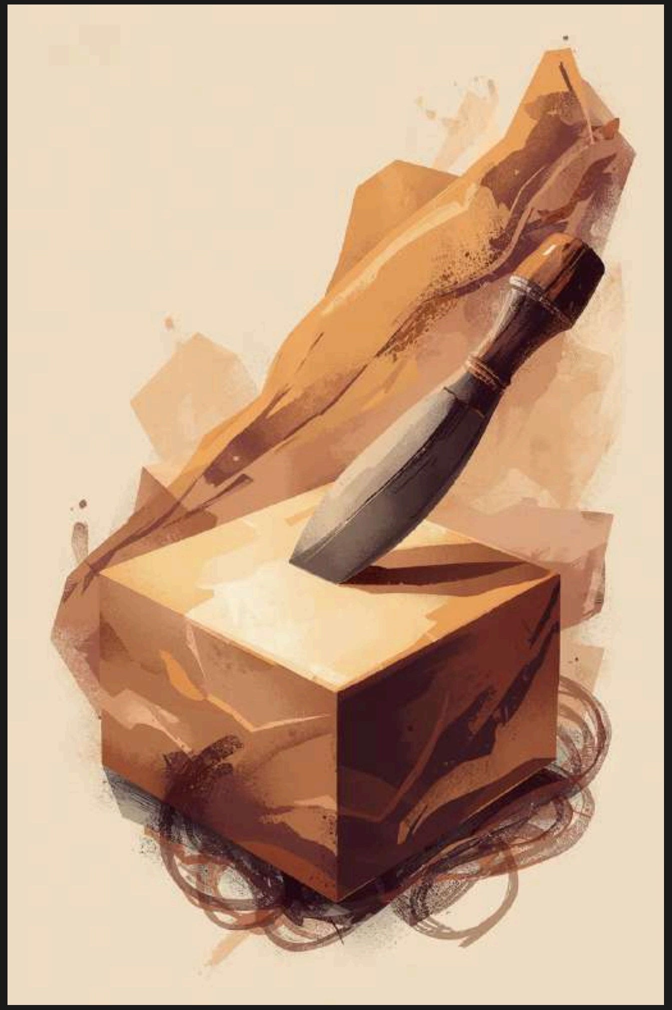
The Big Question

Exploring the impact of specialized agents on system intelligence and performance

This research investigates whether **increasing the number of specialized agents** in a problem-solving system enhances its overall intelligence. By analyzing interactions and performance across various configurations, we aim to uncover the potential benefits and drawbacks of specialization in automated problem-solving contexts.







Formalise

Define clear parameters for problem-solving.

Model

Create structured representations of challenges.

Validate

Ensure solutions meet established criteria.

Refine

Improve solutions through iterative adjustments.

Explain

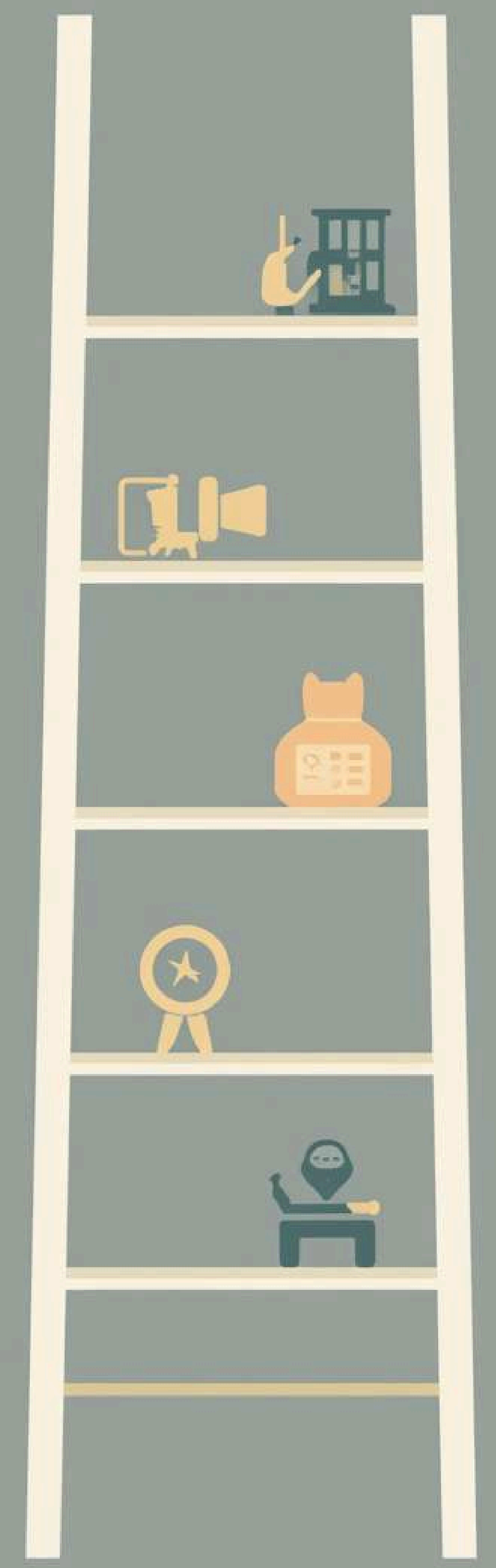
Communicate insights clearly to stakeholders.

The Five Behaviours of Problem Solving

The Delegation Ladder

Understanding the evolution from a single agent to a specialized team of agents

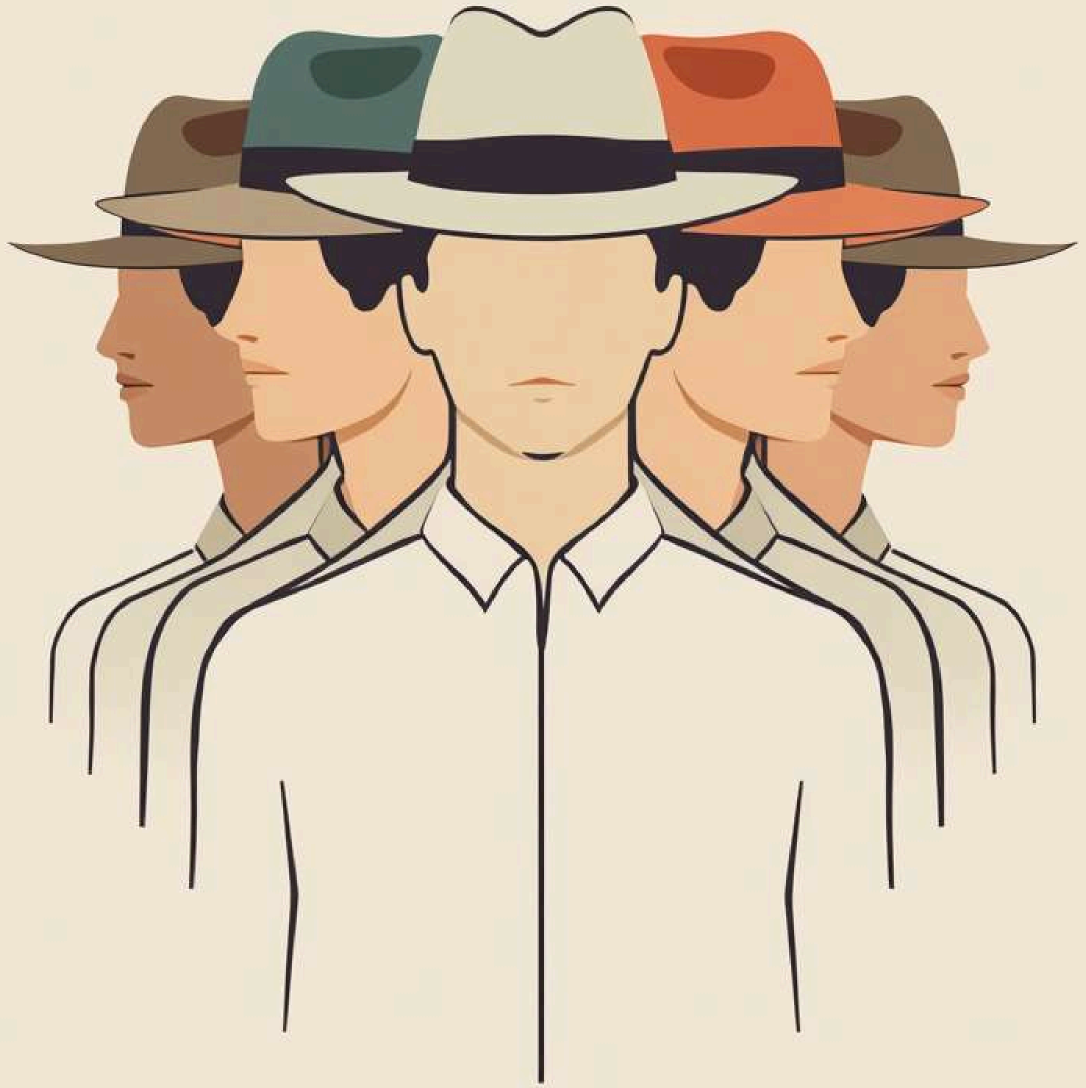
In a complex system, delegation is key for efficiency. The **Delegation Ladder** illustrates the progression from a single agent managing all tasks to multiple agents, each specializing in a specific role. This strategic distribution of responsibilities can enhance performance and effectiveness in tackling combinatorial problems.



The Monolith

One Agent Handles All Five Behaviours in a Unified Approach

In this model, a single agent is responsible for **performing all tasks** required in the combinatorial problem-solving process. This approach simplifies management and communication but may lead to limitations in efficiency and effectiveness. It represents a foundational structure before exploring more complex delegation strategies.



The Specialist Team

Four agents each dedicated to a single role in the system

In this model, the workload is divided among four agents, each specializing in a specific function. This delegation allows for **efficient collaboration** and effective problem-solving, as each agent can focus on their strength. By harnessing individual expertise, the team can achieve greater overall performance and adapt more effectively to challenges.

Experimental Setup

2023

Identified key combinatorial problems

2024

Selected LLMs for evaluation

2025

Implemented delegation levels framework

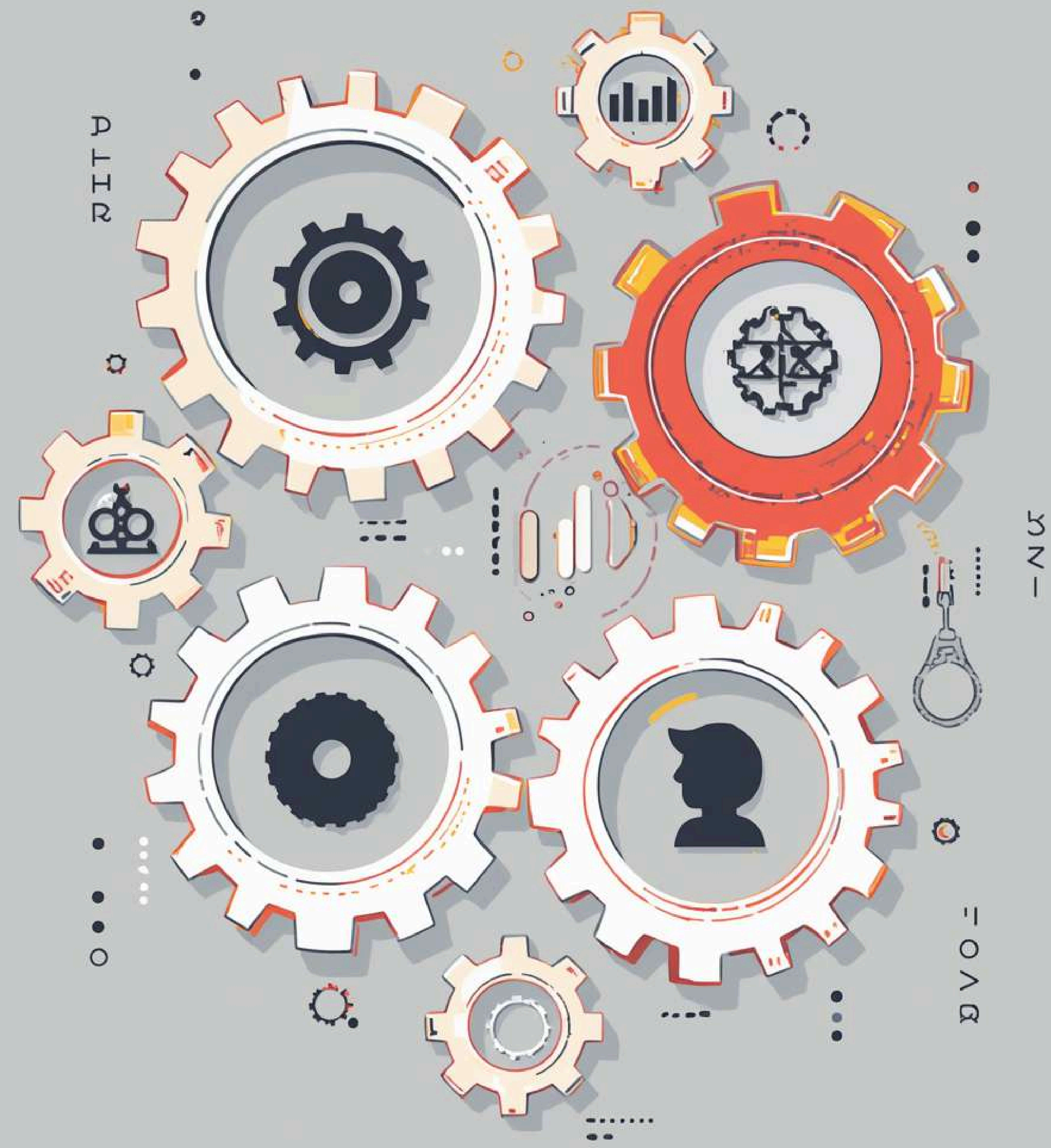
2026

Conducted 120 experimental runs

Pre Registered Prediction

Anticipating the impact of deeper specialization on outcomes

Our hypothesis suggested that increasing the number of specialized agents would lead to improved performance in solving combinatorial problems. This expectation was based on the premise that specialization enhances efficiency, allowing each agent to focus on their strengths, thereby optimizing the overall system's effectiveness.



66%

Final success rate achieved

100%

Initial success rate observed

34% drop

Decline in performance noted

Decline Insights

Easy problems solved; hard problems exposed underlying issues.

The analysis revealed that while **simple tasks** were consistently addressed, it was the **complex challenges** that uncovered flaws in our approach. By examining these discrepancies, we gain a clearer understanding of where our methodology faltered, providing valuable lessons for future implementations.

Mechanisms of Decline

Understanding the Factors Contributing to Performance Decrease in Our Model

In exploring the decline in success rates, we identified **five key mechanisms**. Each mechanism reveals how increasing complexity and redundancy among agents can negatively impact the overall efficiency and effectiveness of our problem-solving approach, leading to unexpected results that challenge initial assumptions.

Errors Accumulate

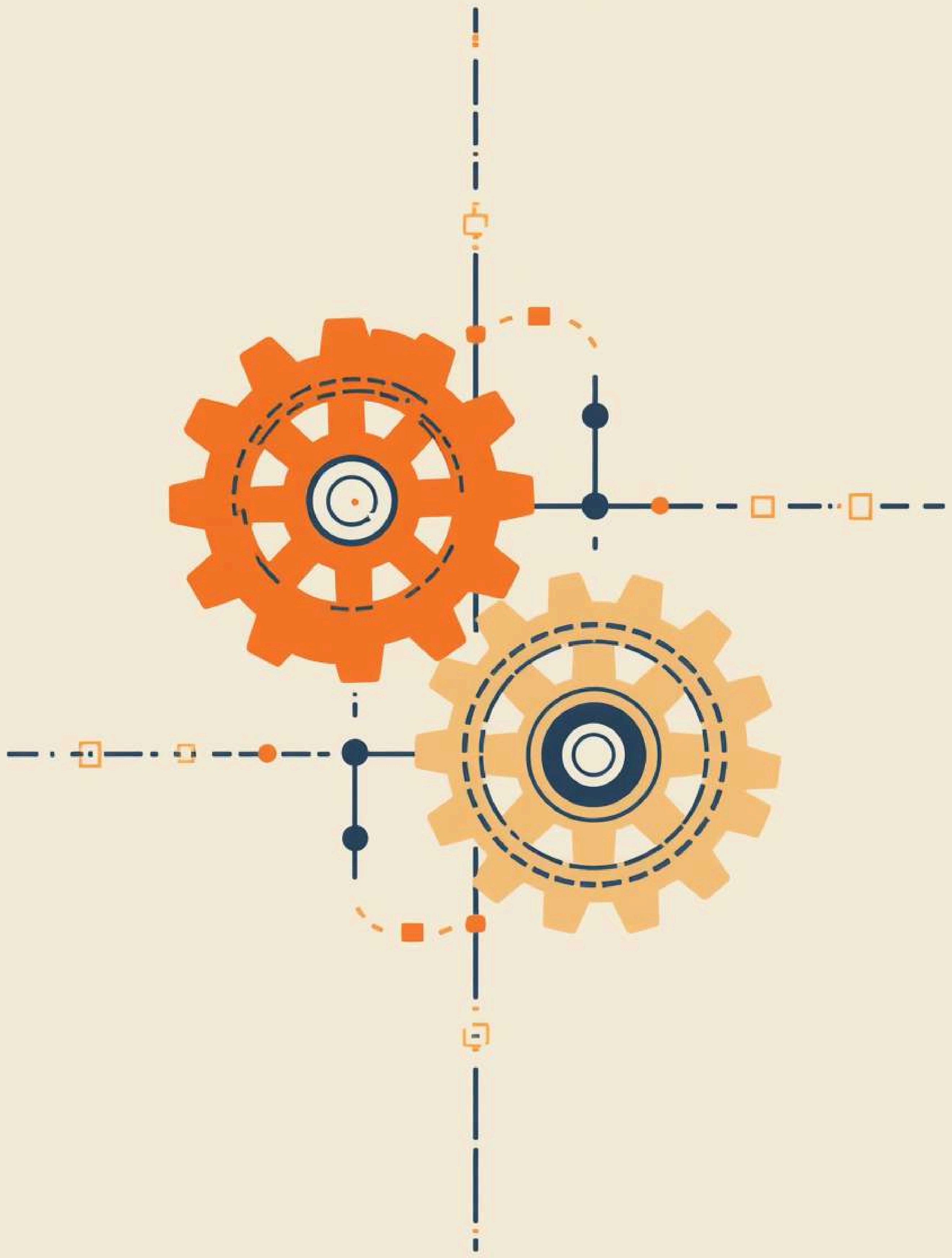
Reliability decreases in multi-stage processes, affecting end-to-end outcomes.

In complex systems, each stage contributes to overall reliability. When each stage operates at **95% reliability**, the cumulative effect significantly reduces the final success rate to only **81%**. This highlights the critical importance of maintaining high reliability across all stages in order to achieve effective outcomes.

Redundancy Concerns

The solver's correctness is compromised by unnecessary validation layers.

Introducing multiple agents to verify correctness can lead to **conflicting assessments**. Since the solver already determines the accuracy of the solution, adding an additional checker may introduce **noise**, reducing overall efficiency and possibly leading to incorrect conclusions. This raises concerns about the effectiveness of multi-agent configurations in this context.



Critic's Impact

The challenges of relying on validators in complex systems

Relying on a validator to identify defects often backfires; they may introduce **additional bugs** rather than providing accurate assessments. This paradox highlights the inherent challenges of using validation as a safety net, emphasizing that a validator's **increased scrutiny** does not guarantee better outcomes in complex problem-solving environments.

Blind Retry Budget

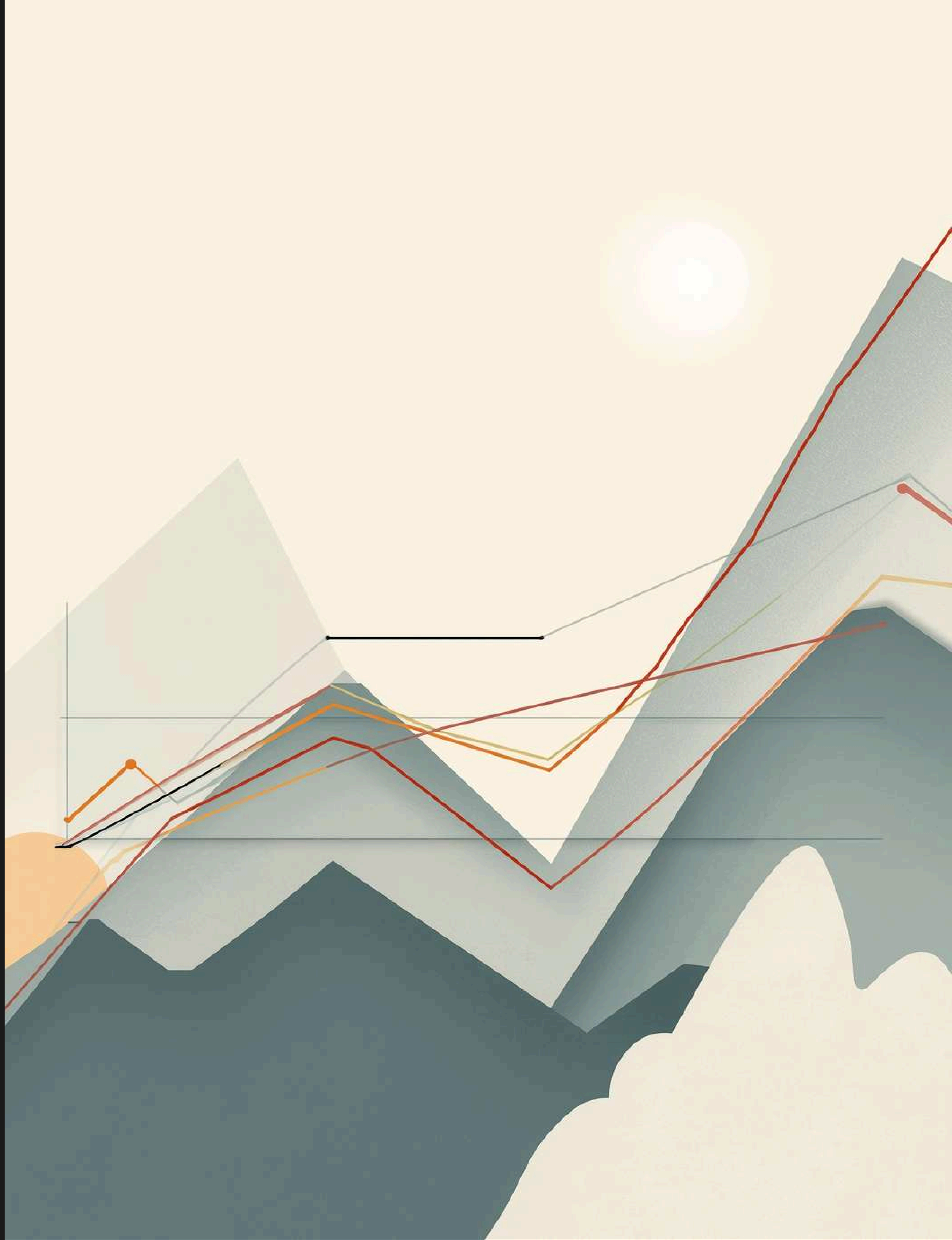
The cost of retries escalates dramatically with increased agent specialization.

In our study, we observed that **three attempts** to solve a problem at Level 3 resulted in significantly higher costs compared to Level 0. This highlights how adding more agents can complicate the solving process, leading to diminishing returns in efficiency and effectiveness across complex combinatorial problems.

Model Performance Variability

Understanding how different models perform at various delegation levels is crucial for success

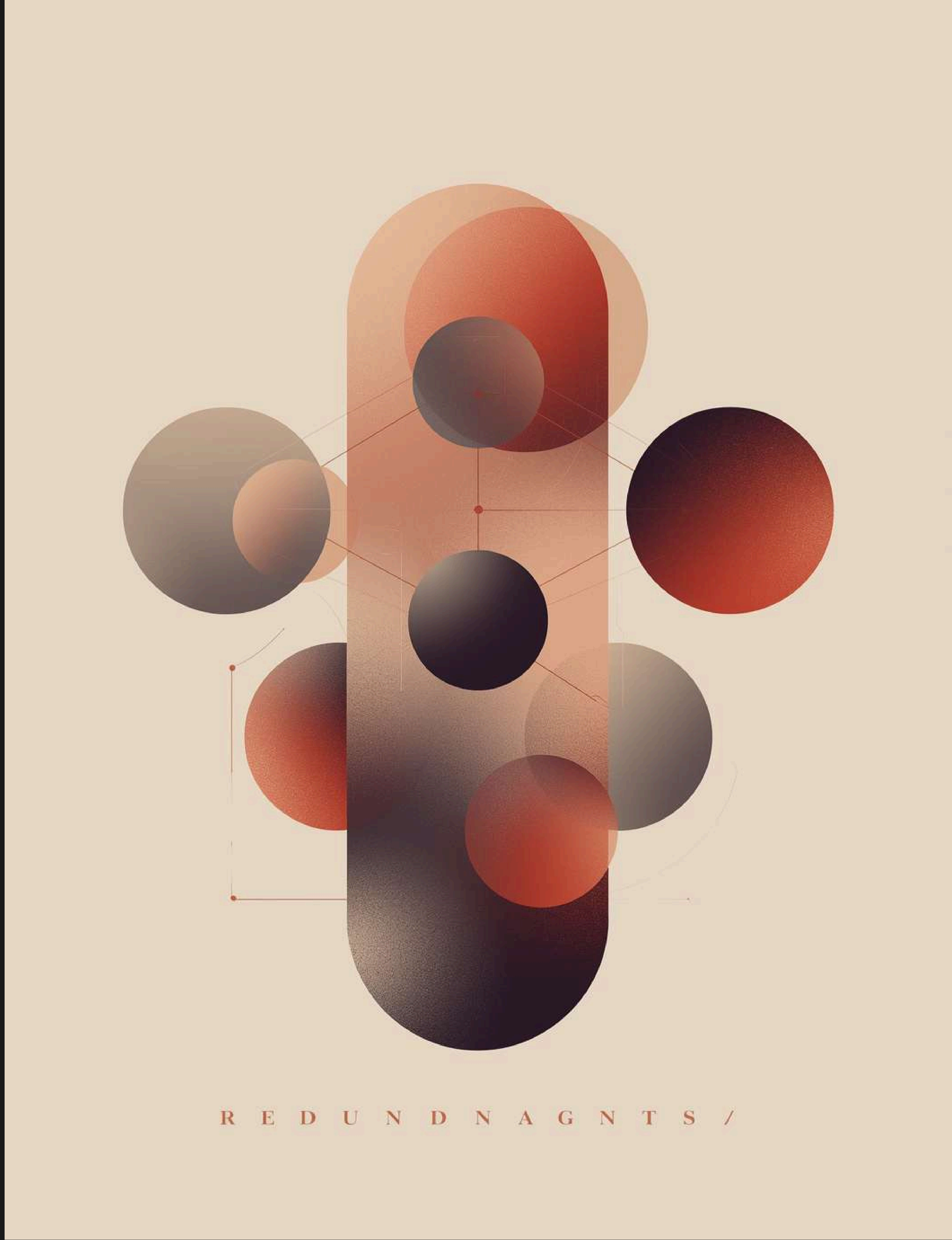
The performance of models can vary significantly depending on the **delegation level.** A model that excels at Level 0 might struggle at Level 3 due to increased complexity or noise introduced by additional agents. This variability emphasizes the importance of choosing the right model for the specific problem context.



The Unifying Principle

Understanding the Impact of Verifier Redundancy on System Efficiency

The **verifier redundancy tax** highlights how employing multiple agents to validate solutions can lead to decreased efficiency. When an exact checker already exists, adding more agents often introduces unnecessary noise and confusion, ultimately hindering the system's performance rather than enhancing it, revealing the balance required in multi-agent systems.



When Multi-Agent Helps

Decomposition in tasks without external checkers can enhance performance and efficiency

In scenarios where an **exact checker is absent**, using multiple agents allows for task decomposition. This strategy can distribute workload and leverage diverse strengths, leading to improved solutions. By segmenting complex tasks, the system can better adapt and respond to unique challenges that arise in problem-solving.

The Noise Factor

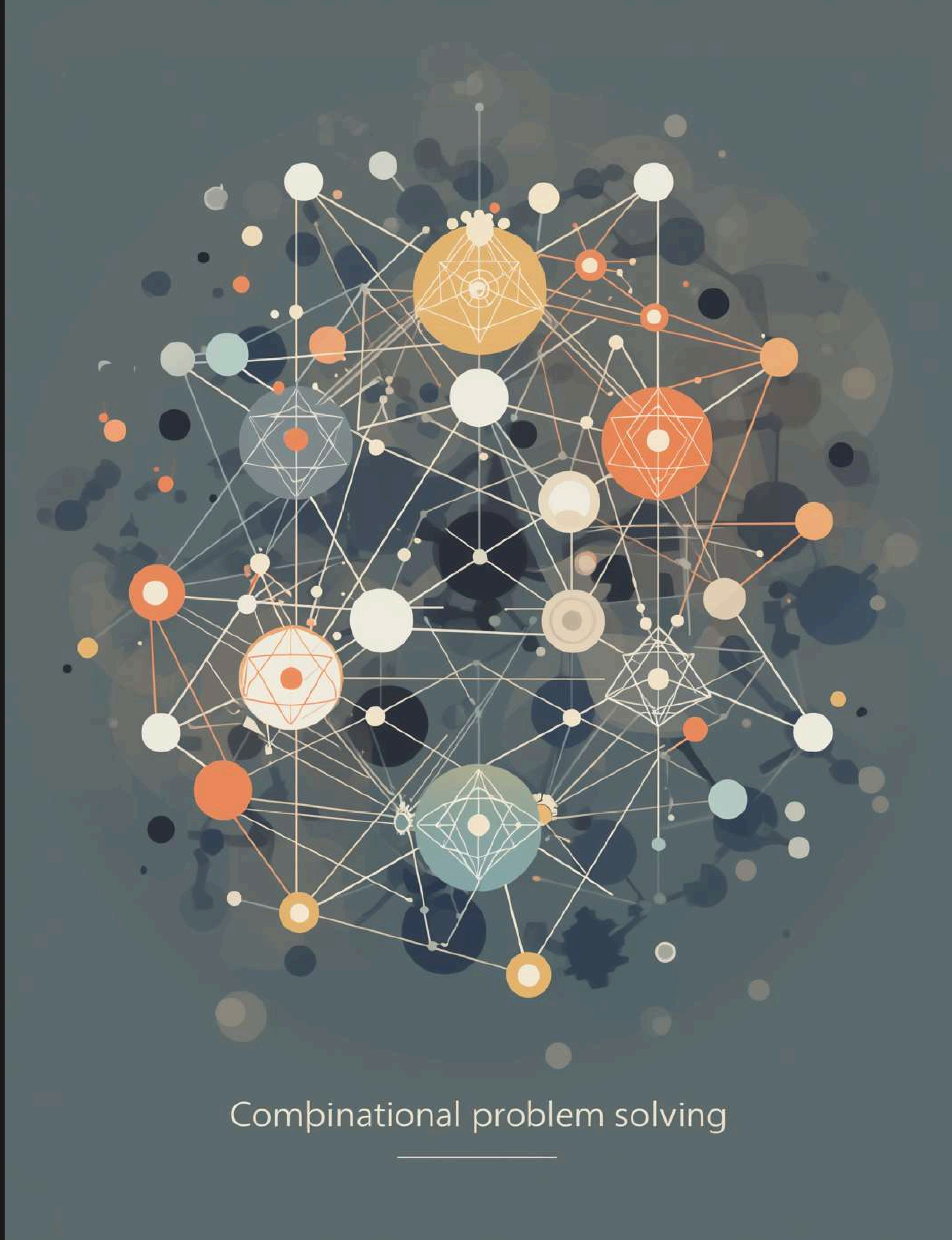
Additional agents can complicate the solving process when exact checkers are present.

Introducing more specialized agents into a system with an **exact checker** can create unnecessary complexity. Instead of enhancing the solution's accuracy, these agents may produce conflicting information, resulting in noise that obscures the correct results. This phenomenon highlights the importance of a balanced approach in agent design.

Clarifying Scope

Previous research like AutoGen and MetaGPT focuses on tasks lacking exact checkers.

This study **restricts rather than contradicts** their findings, emphasizing the importance of exact verification mechanisms. By exploring the interactions in settings with precise checks, we provide a deeper understanding of multi-agent systems and their limitations in complex combinatorial problem-solving contexts.



Limitations of the Study

Key constraints affecting the findings of our research on multi-agent systems

This study faces notable **limitations**: a small sample of hard-tier problems, the exclusive use of sequential topology, and some metrics that remain structurally undefined. These factors may limit generalizability and reliability, suggesting caution when interpreting the results and encouraging further research pathways to address these issues.

Future Directions

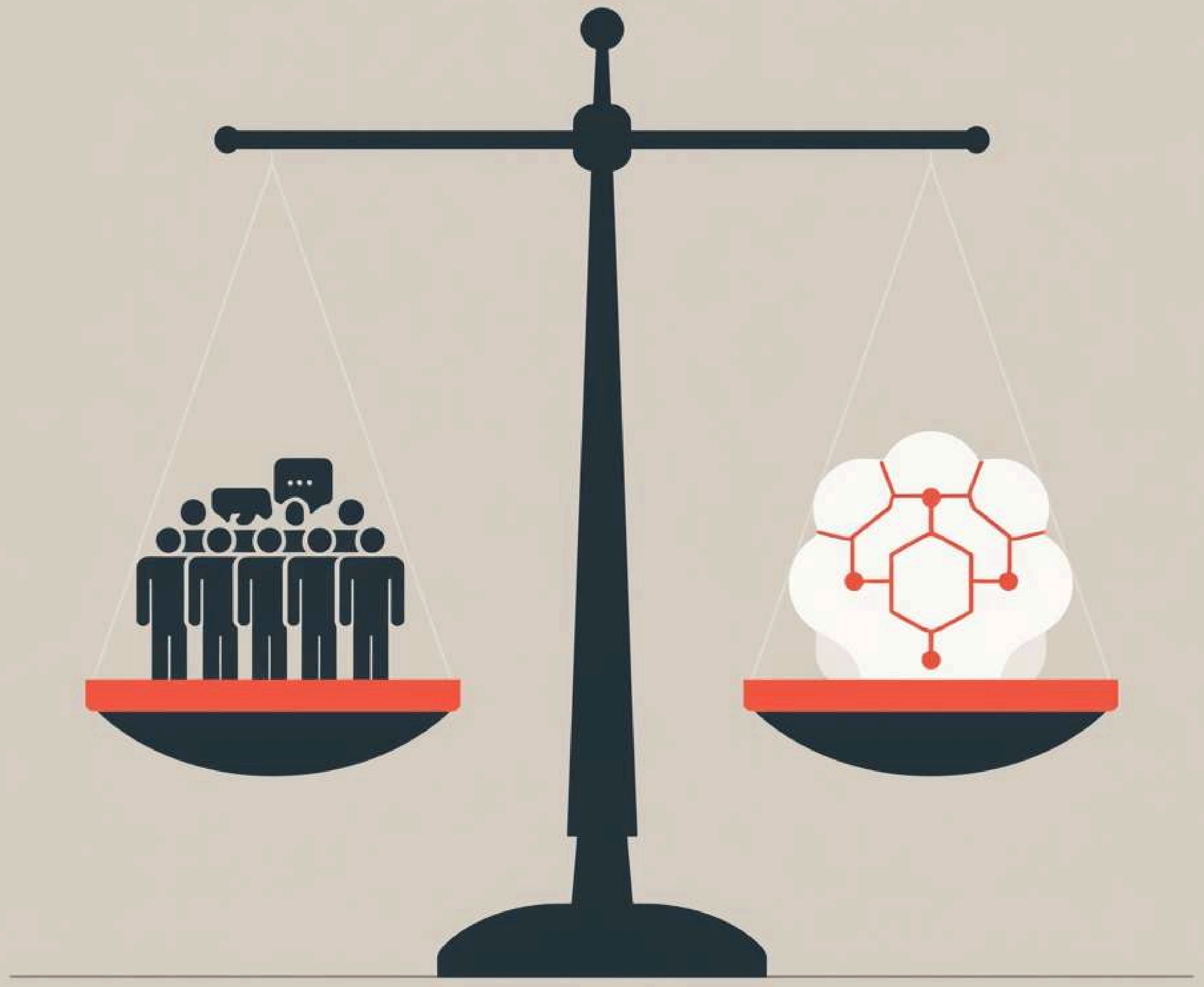
Exploring New Approaches in Agent Delegation and Design Strategies

To enhance our system's efficiency, we propose **reversing the delegation order**, allowing validators to abstain from decisions. Furthermore, exploring **non-sequential designs** could transform how agents interact, potentially leading to improved outcomes and greater adaptability in solving complex combinatorial problems using LLMs and constraint programming.

Take Home Message

Understanding the Role of Architecture in Agent-Based Systems

In conclusion, **more agents** do not necessarily enhance performance. The success of a system relies on its **architecture** and whether it allows for **external checkability**. Careful consideration must be given to the balance between specialization and the complexity introduced by multiple agents to optimize outcomes.



Thank You for Your Attention and Questions